



0812

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Date :



Proposal – Experimental direction

❖ Edge selective early exit on scale or layer

❖ Methods

1. Detect edge with sobel filter/Laplacian filter (not seen when survey paper)
 - Advantage: Extremely fast & Hardware friendly
 - Disadvantage: Lack of semantic understanding
2. Detect edge by training a small MLP (without finetuning model)
 - Advantage: Semantic-aware
 - Disadvantage: Need more computation resource & May need to finetune model

❖ Challenges

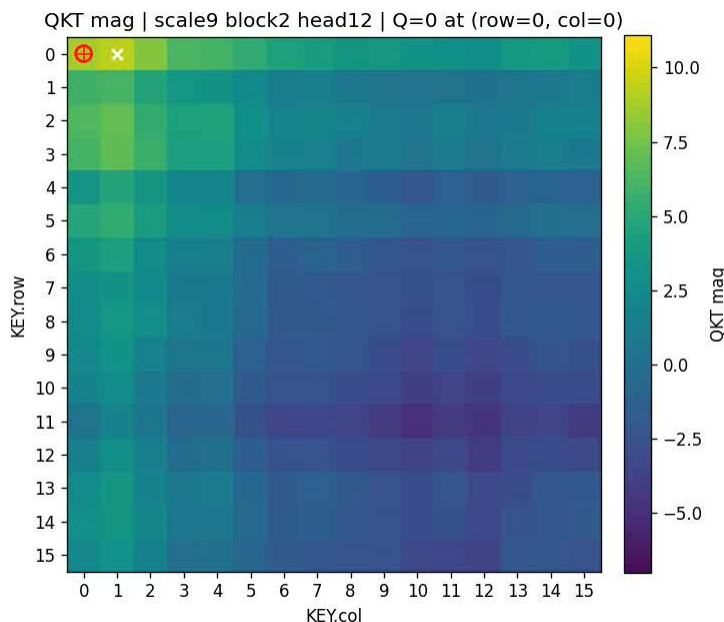
1. KV hole & attention reuse:
Exited token form KV hole, other tokens cannot compute attention
Solution: Make use of the multi-diagonal feature on attention map



Proposal – Experimental direction

❖ High-Scale Local Window Attention

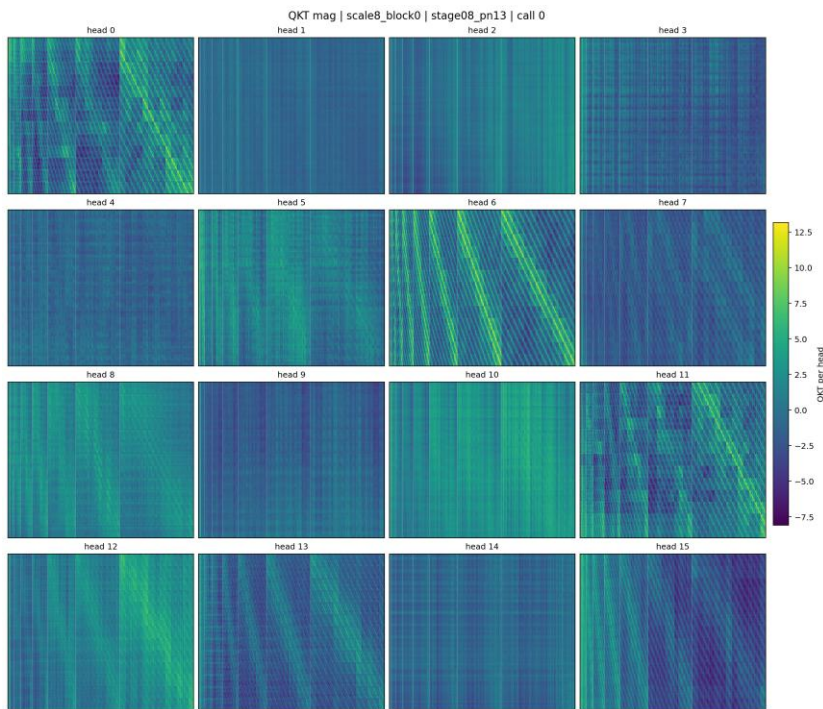
- ❖ Different from article generation, image generation doesn't need full attention. Instead, it probably only need to check attention on a specific region. (Similar work: LiteVAR)



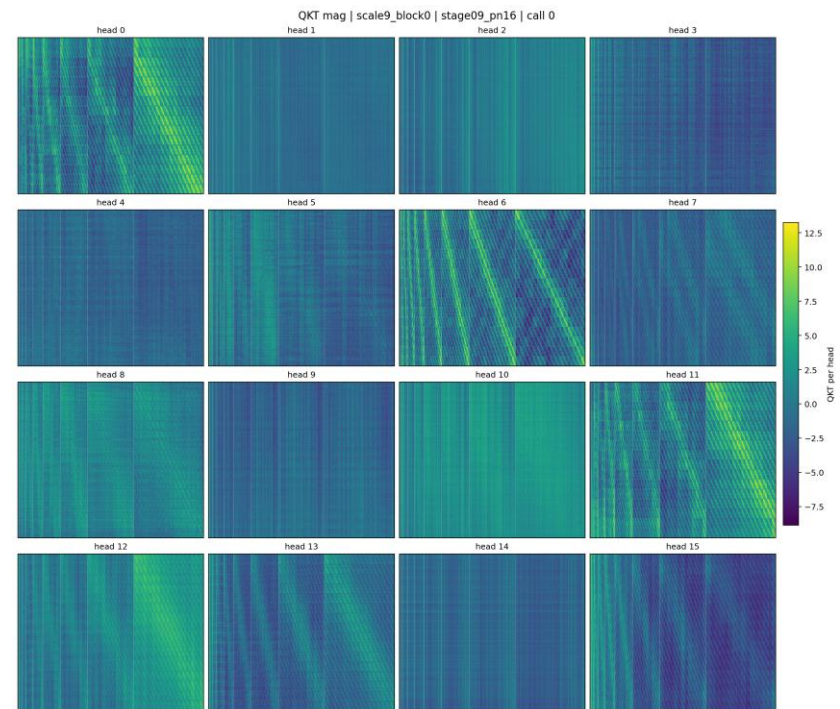


Experiments

❖ QKT heatmap at each scale/block/head



scale8 block0

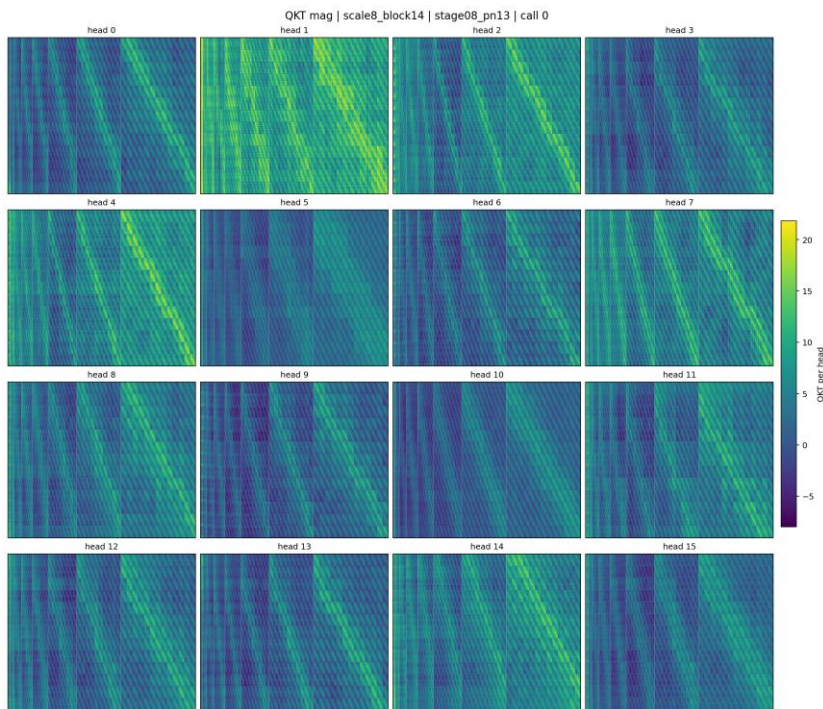


scale9 block0

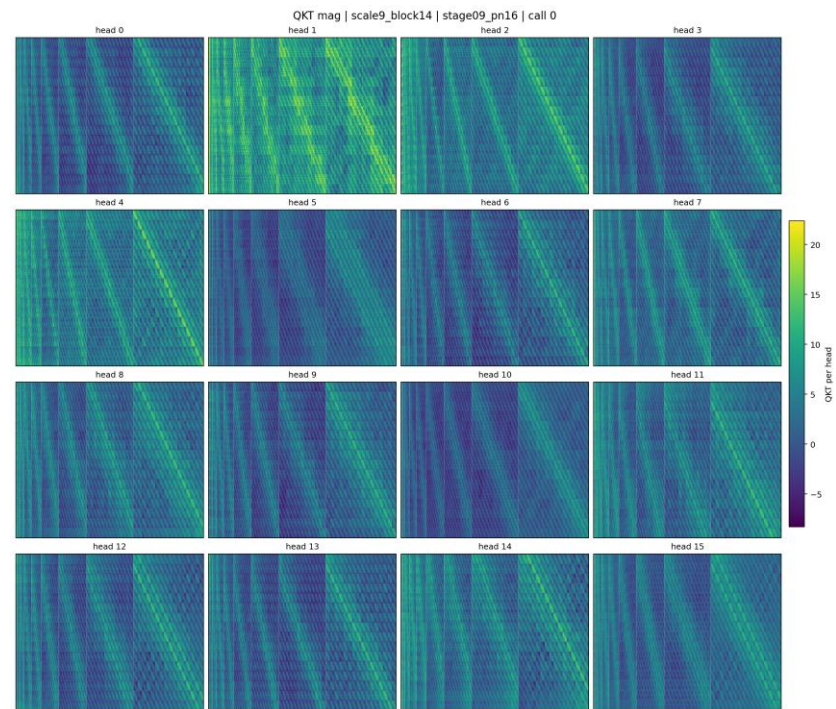


Experiments

❖ QKT heatmap at each scale/block/head



scale8 block14



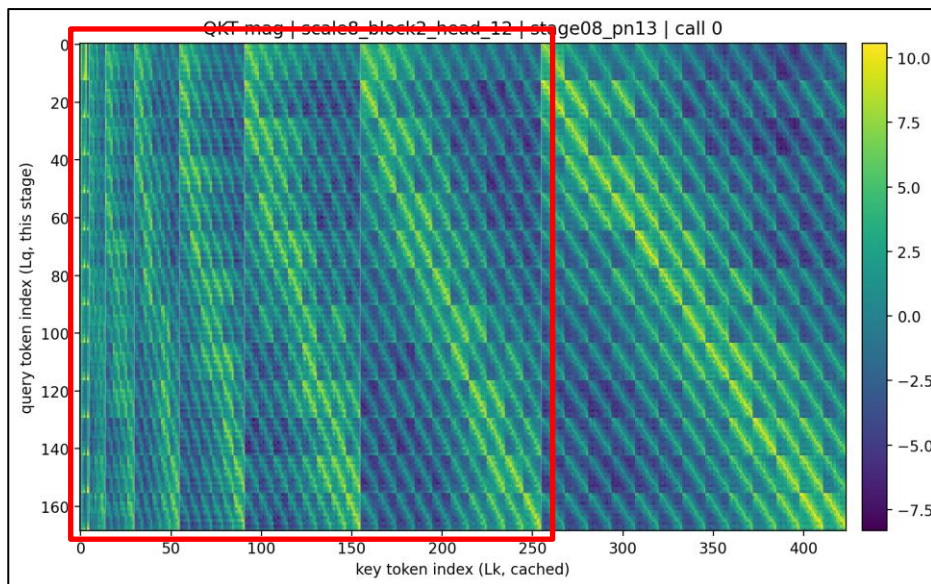
scale9 block14



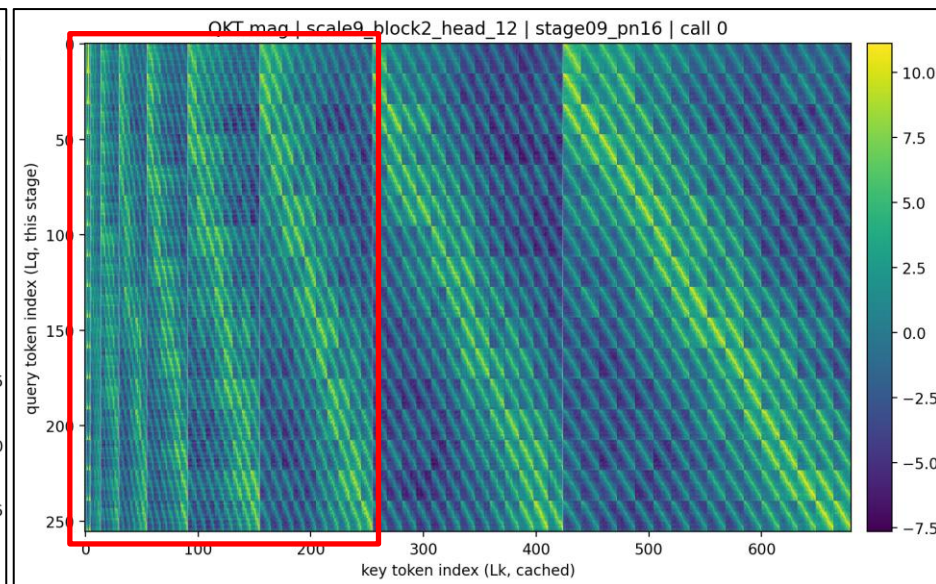
Experiments

Compare $Q[:,K[s0:s7]]$ on scale8&9

And assume that for each head, there's a fixed scale 'a'



Scale8 block2 head12



Scale9 block2 head12

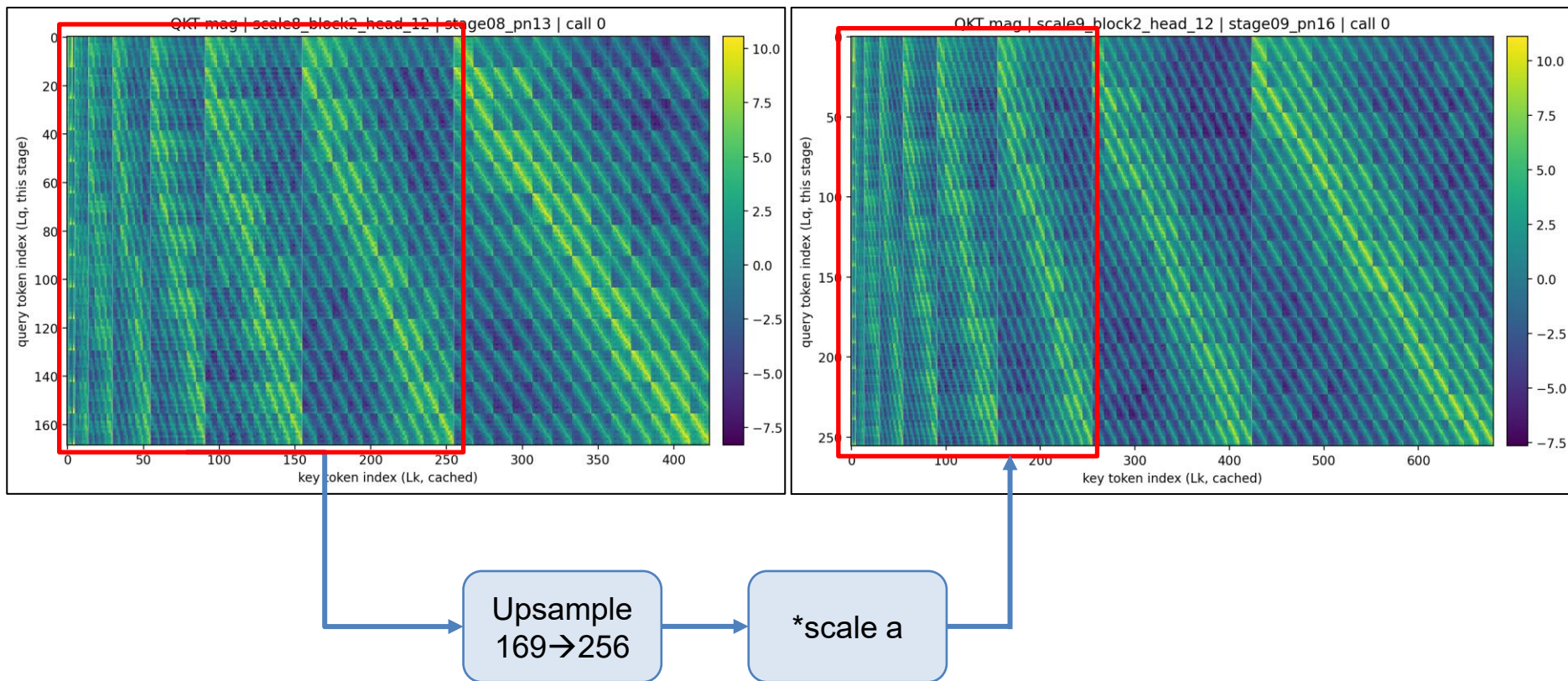
Is it reasonable to reuse attention from prior scale?



Experiments

Compare $Q[:,K[s0:s7]]$ on scale8&9

And assume that for each head, there's a fixed scale 'a'

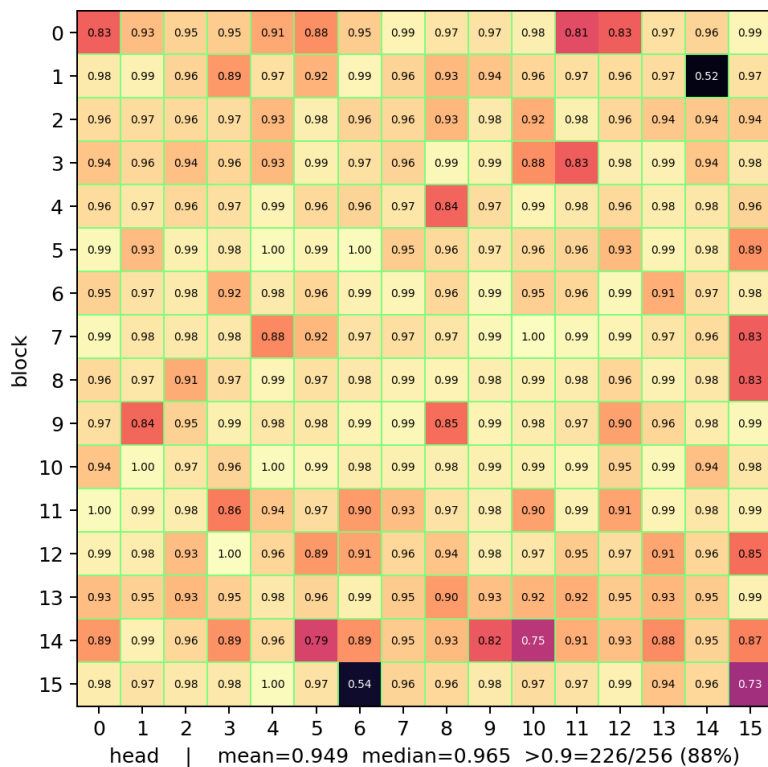




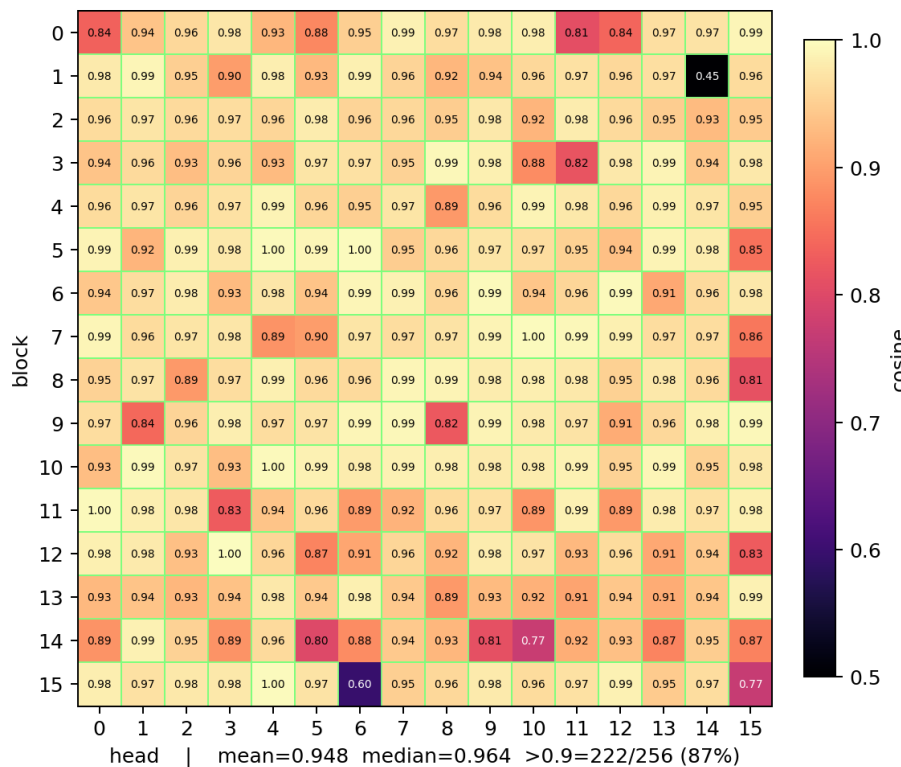
Experiments

s8 → s9 only keys 0-254 (s0-s7) | flattened bilinear upsample cosine

class980 cosine(upsample 169×255 → 256×255 vs QKT9 s0-s7)
outline = cosine > 0.9



class437 same metric
outline = cosine > 0.9





Experiment

